

① Each point we label as $P_i = (x_i, y_i, z_i) \in \mathbb{Z}^3$
for $\{i, j\}$ $i = 1, \dots, 9$

look at the parity points

$$(x_i \bmod 2, y_i \bmod 2, z_i \bmod 2) \in \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$$

There are only $2^3 = 8$ possible triples

$$(0, 0, 0) \quad (1, 1, 1)$$

$$(0, 0, 1) \quad (1, 1, 0)$$

$$(0, 1, 0) \quad (1, 0, 1)$$

$$(1, 0, 0) \quad (0, 1, 1)$$

Idea: Why have I used the parity-class idea?

Well, its to take an abstract value (x_i, y_i, z_i)
and relate it to something we can easily use
and classify them.

Note we have 8 unique points but have
9 given. Hence two distinct points will have
the same value.

For some $i, j \in \{1, 2, \dots, 9\}$

$$(x_i \bmod 2, y_i \bmod 2, z_i \bmod 2) = (x_j \bmod 2, y_j \bmod 2, z_j \bmod 2)$$

Note that $x_i - x_j$, $y_i - y_j$ and $z_i - z_j$ are all even

example:

$$1 \bmod 2 = 1$$

$$2 \bmod 2 = 0$$

$$3 \bmod 2 = 1$$

$$4 \bmod 2 = 0 \quad \text{etc...}$$

:

Now if two numbers are the same $\bmod 2$ then
they are both either even or both odd

ex: $10 \bmod 2 = 0$ } both even
 $8 \bmod 2 = 0$

$11 \bmod 2 = 1$ } both odd

$$a \bmod(2) = 1 \quad b \bmod(2) = 1$$

now their difference is even in either case
even: let $a = 2m$ $b = 2n$, $m, n \in \mathbb{Z}$
 then $a - b = 2(m - n)$

odd: $a = 2m + 1$, $b = 2n + 1$, $m, n \in \mathbb{Z}$
 $a - b = 2m + 1 - (2n + 1) = 2(m - n)$

thus

$$(x_i - x_j, y_i - y_j, z_i - z_j)$$

is a point of even numbers, hence the
 point

$$Q = \left(\frac{x_i - x_j}{2}, \frac{y_i - y_j}{2}, \frac{z_i - z_j}{2} \right)$$

is a point between P_i and P_j which consists
 of integers.

